Semih Utku Canverdi

sutkucanverdi | semihutkucanverdi | utkucnvrdi@gmail.com

Summary

4th-year Software Engineering student focused on game development, algorithmic problem-solving, and AI-driven systems. Passionate about designing core gameplay mechanics, pathfinding logic, and clean software architectures using Unity, C#, and Python.

Projects

Game Development & AI (MağaraJam) *Game Jam Project*
Developed a playable game prototype under tight time constraints using Unity and C#; implemented AI agents utilizing search, pathfinding, and optimization algorithms.

Knight Path Finder GitHub
Designed an algorithm visualizer focusing on pathfinding logic, graph traversal, and interactive problem-solving.

YatDice *Personal Project*
Designed and developed an interactive dice game application from scratch using C#, implementing clean object-oriented architecture, randomized mechanics, structured state management, and robust user interaction.

Air Quality Prediction *University Project*
Built a predictive machine learning model utilizing Python to analyze environmental datasets, perform exploratory data analysis, and execute complete data science workflows.

Work Experience

VISTA Lab (Volunteer) September 2026 - Present
Developing a deep learning model to predict optimal finger positioning for classical piano scores, focusing on data preprocessing, feature extraction, and neural network pipelines.

TEİAŞ July 2026 - August 2026
Developed internal web applications using Java and Angular, contributing to backend RESTful API design and database integration.

TeamSec July 2025 - January 2026
Built dashboards and backend services using FastAPI and Streamlit. Migrated legacy systems to React, analyzed data via Power BI, and optimized SQL queries.

Certificates

May 2026 **Certificate of Achievement - HEALTH-AI Project** (Erasmus+ / European Union Funded)

Education

2023 - June 2027 (expected)	Software Engineering at **Çankaya University**
2017 - 2021	Music/Piano at **Süleymanpaşa Municipality Conservatory**
2019 - 2023	High School at **Namık Kemal Anadolu Lisesi**

Skills

Programming	Python, Java, C/C++, C#, JavaScript, SQL, HTML/CSS
Frameworks	PyTorch, Spring Boot, React.js, FastAPI, Streamlit
Tools	PostgreSQL, Docker, Git, Power BI, Unity
Concepts	Machine Learning, System Architecture, OOP, Data Structures, Pathfinding & Search Algorithms